

Beyond Guam: the cyanobacteria/BMAA hypothesis of the cause of ALS and other neurodegenerative diseases.

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Excitement about neurogenetics in the last two decades has diverted attention from environmental causes of sporadic ALS. Fifty years ago endemic foci of ALS with a frequency one hundred times that in the rest of the world attracted attention since they offered the possibility of finding the cause for non-endemic ALS throughout the world. Research on Guam suggested that ALS, Parkinson's disease and dementia (the ALS/PDC complex) was due to a neurotoxic non-protein amino acid, beta-methylamino-L-alanine (BMAA), in the seeds of the cycad *Cycas micronesica*. Recent discoveries that found that BMAA is produced by symbiotic cyanobacteria within specialized roots of the cycads; that the concentration of protein-bound BMAA is up to a hundred-fold greater than free BMAA in the seeds and flour; that various animals forage on the seeds (flying foxes, pigs, deer), leading to biomagnification up the food chain in Guam; and that protein-bound BMAA occurs in the brains of Guamanians dying of ALS/PDC (average concentration 627 microg/g, 5 mM) but not in control brains have rekindled interest in BMAA as a possible trigger for Guamanian ALS/PDC. Perhaps most intriguing is the finding that BMAA is present in brain tissues of North American patients who had died of Alzheimer's disease (average concentration 95 microg/g, 0.8mM); this suggests a possible etiological role for BMAA in non-Guamanian neurodegenerative diseases. Cyanobacteria are ubiquitous throughout the world, so it is possible that all humans are exposed to low amounts of cyanobacterial BMAA, that protein-bound BMAA in human brains is a reservoir for chronic neurotoxicity, and that cyanobacterial BMAA is a major cause of progressive neurodegenerative diseases including ALS worldwide. Though Montine et al., using different HPLC method and assay techniques from those used by Cox and colleagues, were unable to reproduce the findings of Murch et al., Mash and colleagues using the original techniques of Murch et al. have recently confirmed the presence of protein-bound BMAA in the brains of North American patients dying with ALS and Alzheimer's disease (concentrations >100 microg/g) but not in the brains of non-neurological controls or Huntington's disease. We hypothesize that individuals who develop neurodegenerations may have a genetic susceptibility because of inability to prevent BMAA accumulation in brain proteins and that the particular pattern of neurodegeneration that develops depends on the polygenic background of the individual.